

Disproof of conjecture 00000004515: the planar random Euclidean MST has no $(\log n)^{1/4}$ factor

Jackmeson1

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1 The conjecture and the readings refuted

Statement (English, verbatim). Definition: The random Euclidean MST is the minimum spanning tree on iid points of the unit region. Conjecture: In dimension two the expected total weight is a constant times $\sqrt{n}$ times the $1/4$ power of $\log n$, the exponent $1/4$ is exact and determined by the long-edge density formula. (log-power exponent $1/4$ of the planar MST) (The Chinese text states the same.)

Objects. Let μ be a probability law on $\mathbb{R}^2$ (the law of each point) and let $X_1, \dots, X_n$ be iid with law μ , i.e. $(X_1, \dots, X_n)$ has law $\mu^{\otimes n}$. W_n is the total Euclidean length of the minimum spanning tree of the complete graph on the points, and $E_n = \mathbb{E}[W_n]$. Write $f(n) = \sqrt{n}(\log n)^{1/4}$ ($\log$ is the natural logarithm; another base only changes the constant).

Readings. We refute each of the following, for *every* law μ concentrated on a bounded planar set (in particular the uniform law on the unit square or the unit disk):

- (1) $E_n \sim c f(n)$, i.e. $E_n/(c f(n)) \rightarrow 1$, for any real c ;
- (2) $E_n = c f(n)$ for all large n , for any $c > 0$;
- (3) $E_n = \Theta(f(n))$ (“the exponent $1/4$ is exact”): $c_1 f(n) \leq E_n \leq c_2 f(n)$ eventually, with $c_1, c_2 > 0$;
- (4) $E_n \geq c f(n)$ for infinitely many n , for any $c > 0$.

Reading (4) is the weakest; (1) with $c > 0$, (2) and (3) each imply it. The clause “determined by the long-edge density formula” only explains where the exponent comes from; it falls with the exponent. *Not covered:* unbounded regions of unit area, power-weighted edge lengths $|e|^\alpha$ with $\alpha \neq 1$, and $c = 0$ in (2) (for a Dirac law $W_n = 0$; the text’s “exponent $1/4$ is exact” excludes $c = 0$).

2 Definitions (as in the Lean file)

Definition 1. For $p, q \in \mathbb{R} \times \mathbb{R}$, $d(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2}$ (**edist2**; Mathlib’s **dist** on $\mathbb{R} \times \mathbb{R}$ is the sup metric, so the Euclidean formula is written out). For $x : \text{Fin } n \rightarrow \mathbb{R}^2$ and a simple graph G on $\text{Fin } n$, $\text{len}_x(G) = \sum_{\{i, j\} \in E(G)} d(x_i, x_j)$ (**graphLen**). The MST length is

$$W(x) = \min_{T \text{ a tree on } \text{Fin } n} \text{len}_x(T)$$

(**mstLen**, an infimum over the finite set of trees; **mstLen_attained** shows it is attained for $n \geq 1$). Every tree on the vertex set $\text{Fin } n$ is a spanning tree of the complete graph. The expectation

is $E_n(\mu) = \int W d\mu^{\otimes n}$ (`expMST`, with `Measure.pi`); `measurable_mstLen` and `integrable_mstLen` show that it is a genuine finite expectation. The uniform law on the unit square is `Leb|[0,1]2` (`uniformSq`, a probability measure).

The four readings are the Lean predicates `AsympReading`, `EqReading`, `ThetaReading` and `LowerIOReading`, with $f = \text{prof}$.

3 The strip bound

Lemma 2 (`exists_connected_short`, `mstLen_le_unitSq`). *For $n \geq 1$ points $x_1, \dots, x_n \in [0, 1]^2$ there is a connected graph G on them with $\text{len}_x(G) \leq 3\sqrt{n} + 2$. Hence $W(x) \leq 3\sqrt{n} + 2$.*

Proof. Let $k = \lfloor \sqrt{n} \rfloor + 1$, so $\sqrt{n} \leq k \leq \sqrt{n} + 1$. The strip index of a height $y \in [0, 1]$ is $s(y) = \min(\lfloor ky \rfloor, k - 1)$, so $s \leq ky \leq s + 1$ and $0 \leq s \leq k - 1$. Sort the points by the key $s(y) + x/2$ (`Tuple.sort`); since $x/2 \in [0, 1/2]$, this is the lexicographic order (strip, then abscissa). Let $q_0, \dots, q_{n-1}$ be the sorted points and G the path $q_0 q_1 \dots q_{n-1}$ (the image of `pathGraph` under the sorting permutation, so connected). For consecutive $p = q_j$, $q = q_{j+1}$ with strips $s \leq t$: if $s = t$ then $q_1 \geq p_1$, so $|q_1 - p_1| = q_1 - p_1$; if $s < t$ then $|q_1 - p_1| \leq q_1 - p_1 + 2 \leq q_1 - p_1 + 2(t - s)$. Also $|q_2 - p_2| \leq (t - s + 1)/k$. Since $d \leq |\Delta_1| + |\Delta_2|$,

$$d(q_j, q_{j+1}) \leq F(j+1) - F(j), \quad F(j) = (q_j)_1 + (2 + 1/k)s(q_j) + j/k$$

(`step_le`). Every edge of G is such a pair, so telescoping gives

$$\text{len}(G) \leq F(n-1) - F(0) \leq 1 + (2 + 1/k)(k-1) + (n-1)/k \leq 3\sqrt{n} + 2$$

(`strip_arith`: multiply by k and use $k \leq \sqrt{n} + 1$, $n \leq k\sqrt{n}$). A connected graph contains a spanning tree $T \leq G$ (`Mathlib.Connected.exists_isTree_le`) and $\text{len}(T) \leq \text{len}(G)$ since lengths are nonnegative, so $W \leq \text{len}(G)$ (`mstLen_le_of_connected`). $\square$

Lemma 3 (`mstLen_le_bounded`). *If $S \subset \mathbb{R}^2$ is bounded, there is $C \geq 0$ with $W(x) \leq C(3\sqrt{n} + 2)$ for all $n \geq 1$ and all $x \in S^n$.*

Proof. $S \subseteq [-R, R]^2$ for some $R > 0$. The map $\phi(p) = ((p_1 + R)/2R, (p_2 + R)/2R)$ sends S into $[0, 1]^2$ and $d(p, q) = 2R d(\phi p, \phi q)$ (`edist2_sc`). Apply the previous lemma to $\phi(x_i)$ and take $C = 2R$. $\square$

4 The disproof

Theorem 4 (`ratio_tendsto_zero`, `not_frequently_lower`). *Let μ be a probability law such that μ -almost every point lies in a bounded set S . Then $0 \leq E_n \leq C(3\sqrt{n} + 2)$ for $n \geq 1$, and for every real c and every $\beta > 0$,*

$$\frac{E_n}{c\sqrt{n}(\log n)^\beta} \longrightarrow 0.$$

Consequently, for $c > 0$, $E_n \geq c\sqrt{n}(\log n)^\beta$ holds for only finitely many n .

Proof. Almost surely all $X_i \in S$, so $0 \leq W \leq C(3\sqrt{n} + 2)$ a.s. and the bound integrates (`expMST_le`). For $c = 0$ the quotient is 0 by the convention $a/0 = 0$. For $c \neq 0$ and $n \geq 3$, $|E_n/(c\sqrt{n}(\log n)^\beta)| \leq 5C/(|c|(\log n)^\beta) \rightarrow 0$ because $(\log n)^\beta \rightarrow \infty$. If $E_n \geq c\sqrt{n}(\log n)^\beta$ for infinitely many n with $c > 0$, the quotient would be ≥ 1 infinitely often, a contradiction. $\square$

Theorem 5 (`conjecture_4515_false`, `conjecture_4515_false_uniform`). *For every such μ , and in particular for the uniform law on $[0,1]^2$: (1) fails for every real c ; (2) fails for every $c > 0$; (3) fails; (4) fails for every $c > 0$.*

Proof. (1): the quotient tends to $0 \neq 1$. (4): previous theorem with $\beta = 1/4$. (2) and (3) each give $E_n \geq cf(n)$ for all large n , hence for infinitely many n , contradicting (4). $\square$

Context. The retrieved Wikipedia article “Euclidean minimum spanning tree” (https://en.wikipedia.org/wiki/Euclidean_minimum_spanning_tree) states: “The total length of the edges, for points in a unit square, is at most proportional to the square root of the number of points.” The proof above does not use it.

5 Lean formalization

File `lean/Conjecture4515/Basic.lean` (namespace `C4515`, only `import Mathlib`, `Mathlib v4.33.1`). Main statements:

```
theorem conjecture_4515_false (mu : Measure (R x R)) [IsProbabilityMeasure mu]
  {S : Set (R x R)} (hS : Bornology.IsBounded S) (hmu : forall^ae p d mu, p in S) :
  (forall c : R, not AsympReading mu c) /\ (forall c > 0, not EqReading mu c) /\
  not ThetaReading mu /\ (forall c > 0, not LowerIOReading mu c)
theorem conjecture_4515_false_uniform :
  (forall c : R, not AsympReading uniformSq c) /\ (forall c > 0, not EqReading uniformSq c) /\
  not ThetaReading uniformSq /\ (forall c > 0, not LowerIOReading uniformSq c)
```

(ASCII transcription of the Lean statements.) Supporting theorems: `mstLen_le_unitSq`, `mstLen_le_bounded`, `integrable_mstLen`, `expMST_le`, `ratio_tendsto_zero`, `not_frequently_lower`.

Axiom audit. `#print axioms` for both main theorems reports only `propext`, `Classical.choice`, `Quot.sound`. No sorry, no `native_decide`, no added axioms. Built with lake build.